

Quantum Magnetometer - NV^- Sensors

September 22, 2026

Abstract

A nitrogen-vacancy (NV^-) center is a deliberate flaw in a diamond crystal: one carbon replaced by nitrogen, with the neighboring lattice site left empty. The broken bonds around it hold a small set of electronic states that sit inside diamond's wide energy gap, so the defect behaves like a single atom held rigidly in place. Two of its electrons are unpaired, giving the defect a magnetic spin with three settings. Green light prepares that spin in a known setting and also reports it, because the two ways the defect can return to its ground state — one bright, one dark — occur with different probabilities for different spin settings. Microwaves at 2.87 GHz move the spin between settings, and a magnetic field shifts the frequency at which they do so. These notes follow that chain of reasoning in the order it was presented, ending with the measurement itself: a single dip in a fluorescence spectrum splitting into two, separated by $\Delta f = 2\gamma B$. Technicalities of this field and technology were smoothed out, or has been avoided.

Contents

1	The claim: a defective diamond is a very good magnetic sensor	2
2	The defect: breaking the fourfold bonding of diamond	2
3	Why a defect behaves like a trapped atom	3
4	The part that senses the field: three settings, not a continuum	4
5	Green light in, red light out	4
6	The two ways down — the heart of the mechanism	5
7	What the microwaves do	7
8	The graph: fluorescence versus microwave frequency	7
9	Reading the graph: what the position and the width mean	8
9.1	The position: an energy, via $E = hf$	8
9.2	The width: how sharply the energy is defined	8
10	Turning on a magnetic field: one dip becomes two	9
10.1	The same statement in two lines of algebra	9
10.2	Reading a field off the graph	10

11 Loose ends worth knowing	11
11.1 The center of the pattern carries information too	11
11.2 Four orientations, and why that is a feature	11
11.3 Why many NV^- centers instead of one	11
12 What it is used for	11
13 The argument in one page	12

1 The claim: a defective diamond is a very good magnetic sensor

Diamond is the most orderly material we know: carbon atoms locked into a rigid lattice, each bonded to four neighbors, so tightly bound that the crystal is transparent, chemically inert, and mechanically almost immovable. This is great for using them in manufacturing, or in case you drop your ring, but we are looking for something else.

Take that perfect lattice, break it in one specific way, and the resulting flaw behaves like a single atom frozen in place and as if shielded from its surroundings. It will then tell you, by changing its brightness, how strong the magnetic field is at its location — not the field averaged over a room, but the field at a point you can place within a nanometer of whatever you want to measure.

The rest of these notes is the chain of reasoning that runs from “a flaw in a crystal” to “a magnetometer,” one link at a time.

2 The defect: breaking the fourfold bonding of diamond

Every carbon atom in perfect diamond is bonded to four neighbors. That fourfold (tetrahedral) coordination is what makes the crystal so stiff and so transparent (for the pure ones at least). The electrons are strongly bound in the crystal lattice, and diamond has a large electronic band gap. As a result, visible light normally passes through without being absorbed strongly.

The nitrogen-vacancy center is made by spoiling that arrangement at two neighboring sites. Replace one carbon by a *nitrogen* atom, and remove the carbon next door, leaving a *vacancy* — an empty (lattice) site. Figure 1 shows the before and after.

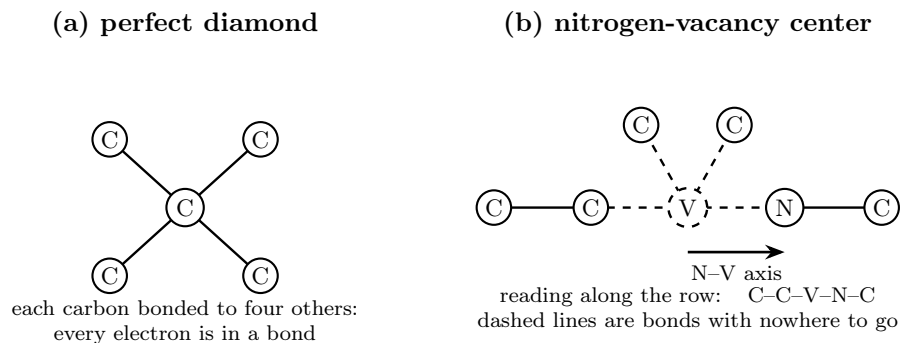


Figure 1: (a) In perfect diamond every carbon sits at the center of a fourfold arrangement of neighbors. (b) The NV^- center: reading along the chain, C-C-V-N-C. The vacancy has four neighbors of its own — three carbons and the nitrogen — and each of them is left holding a bond with nothing to bond to. Those unsatisfied (“dangling”) bonds are where the defect’s own electronic states come from, and the line joining the vacancy to the nitrogen defines the NV^- axis.

3 Why a defect behaves like a trapped atom

Pure diamond is an insulator with an unusually wide energy gap (energy required to enfranchise an electron from its atom, from its valence band), about 5.5 eV separates the electronic states that are full (the valence band) from those that are empty (the conduction band, free to flow like in conductors, “sea of electrons”). No visible photon carries that much energy, which is exactly why diamond is clear — visible light passes through because there is nothing for it to excite.

The dangling bonds around the vacancy change that locally (so a diamond still looks like a diamond from afar). They combine into a small number of *localized* electronic states whose energies land *inside* the forbidden gap, as sketched in Fig. 2. Two features of that picture matter:

- **Localized** means the electrons occupying these states are stuck at the defect, confined to a region about the size of an atom. They cannot roam the crystal.
- **Inside the gap** means visible light can excite them, and that the surrounding diamond — transparent and magnetically quiet — offers them almost nothing else to interact with.

In other words, the NV^- center is an isolated atom that someone has bolted into a rigid, transparent, chemically inert mount. Atomic physicists go to great lengths — vacuum chambers, laser traps — to produce that situation. Here the crystal supplies it for free, at room temperature, in open air.

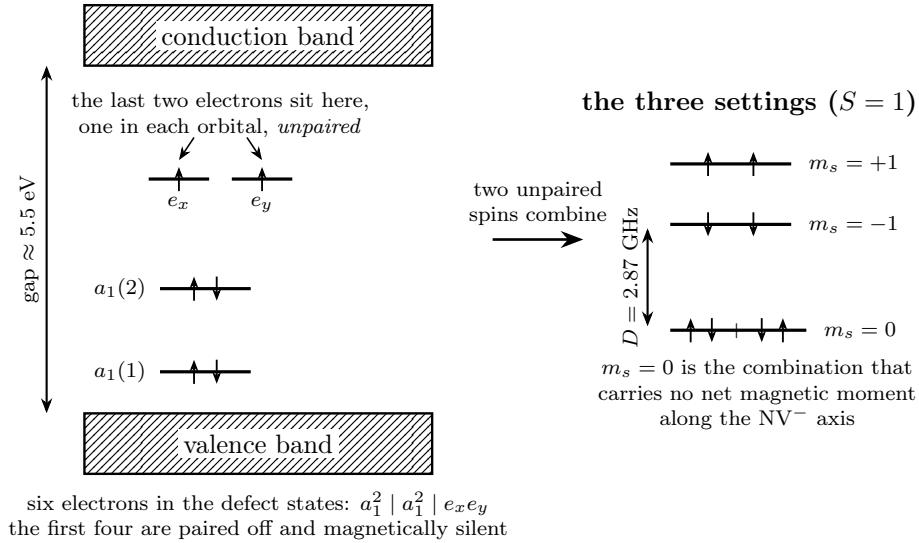


Figure 2: Left: the dangling bonds around the vacancy produce localized states inside diamond’s 5.5 eV gap. Six electrons fill them in the pattern $a_1^2 | a_1^2 | e_x e_y$: the lower two orbitals take paired electrons (one up, one down — magnetically silent), while the last two electrons occupy the degenerate pair e_x, e_y one apiece and stay *unpaired*. Right: two unpaired spins can be arranged three ways — both up, both down, or the symmetric combination of one up and one down — and these are the three settings $m_s = +1, -1, 0$. Even with no magnetic field present, $m_s = 0$ sits lower than $m_s = \pm 1$ by $D = 2.87 \text{ GHz}$.

4 The part that senses the field: three settings, not a continuum

The two unpaired electrons in e_x and e_y are the sensor. Together they act like a tiny magnetic needle attached to the defect — a needle in the same sense that a compass needle is, in that its energy depends on how it is oriented relative to a magnetic field. (Needle of a compass aligns with the Earth’s magnetic field from south to north to minimize its energy due to magnetism the needle exhibits.)

The difference from a compass is that this needle cannot point in arbitrary directions. It has exactly *three* allowed settings, labeled

$$m_s = 0, \quad m_s = +1, \quad m_s = -1,$$

corresponding to the three arrangements on the right of Fig. 2: both spins up, both down, or the balanced combination of one up and one down (a quantum superposition). Think of them as three rungs on a very short ladder. Everything that follows is about (i) working out which rung the defect is on, and (ii) noticing that the spacing between rungs depends on the magnetic field.

Two facts about these rungs carry the whole measurement:

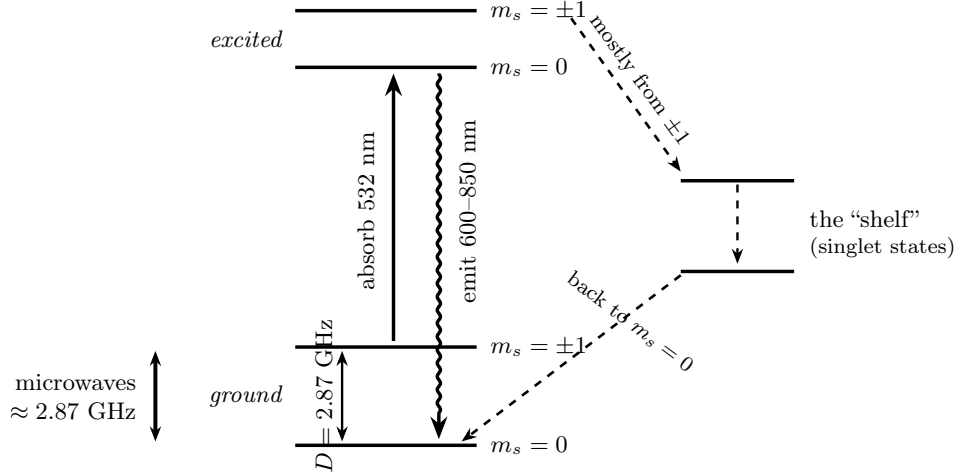
- **Even with no magnetic field anywhere, the rungs are unevenly spaced.** The $m_s = 0$ state sits lower in energy than $m_s = \pm 1$. The two unpaired electrons interact with each other, and that interaction alone holds the settings apart. This built-in gap is the *zero-field splitting*, and it corresponds to a gap of $D = 2.87$ GHz between $m_s = 0$ and $m_s = +1$ (Fig 3 and 4).
- **With no field, $m_s = +1$ and $m_s = -1$ have identical energy.** “Both up” and “both down” are equally good when there is no field to distinguish one direction from the other (left of Fig 4). That is precisely the tie a magnetic field will break — and breaking it is the measurement.

5 Green light in, red light out

Shine green laser light (typically 532 nm) on the diamond and the NV^- center absorbs it, promoting/exciting an electron and carrying the defect from its ground state to an excited state as can be seen in Figure 3. A few nanoseconds later it falls back down and emits a photon of lower energy, in the red and near-infrared band (roughly 600–850 nm).

This is ordinary fluorescence — the same process that makes a highlighter pen glow. Green in, red out, continuously, for as long as the laser is on. A single NV^- center will do this for years without bleaching, which by itself makes NV^- diamond useful as a biological marker.

The point to hold onto is that **the brightness, the intensity of the fluorescence of that red glow is the only thing we ever actually measure.** Every result in this talk is a photodetector watching red light get slightly brighter or slightly dimmer.



Vertical scale is schematic: the optical gap is about 200,000 times larger than the microwave gap.

Figure 3: The working diagram of an NV⁻ center. Solid arrows are optical transitions that emit or absorb a photon; dashed arrows are transitions that do not. Green light drives the center up; it usually falls straight back down and emits a red photon, but occasionally — and much more often if it started in $m_s = \pm 1$ — it takes the detour through the singlet “shelf” and emits nothing we detect. Microwaves at 2.87 GHz move the center between $m_s = 0$ and $m_s = \pm 1$ within the ground state.

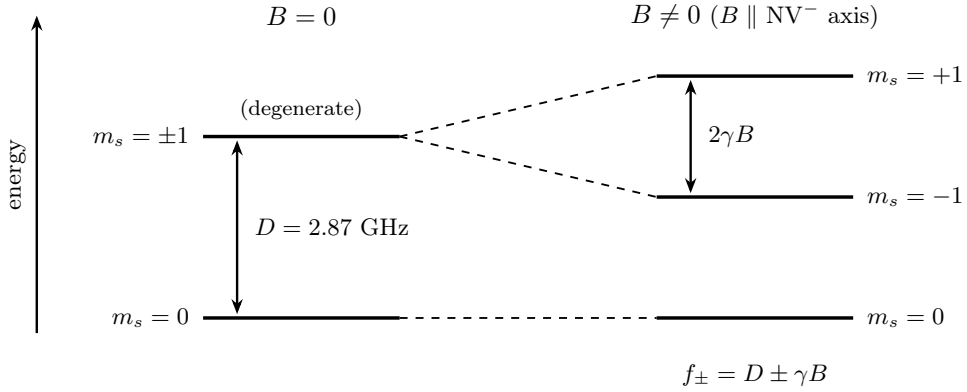


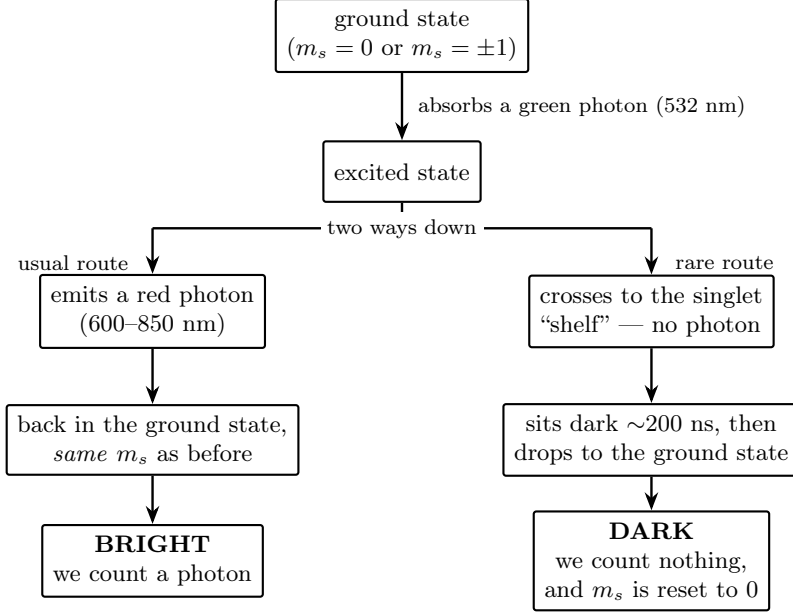
Figure 4: Zero field splitting on the left and splitting of the states after a B field is applied.

6 The two ways down — the heart of the mechanism

Once the NV⁻ center is in the excited state it has a choice of routes back to the bottom, shown as a branching in Fig. 5. This is the single most important idea in the talk, so it is worth stating slowly.

Route 1 — straight down (bright). The center drops directly from the excited state to the ground state and emits a red photon. The spin setting is preserved: whatever rung it was on before, it is on afterwards. We detect the photon. This is the route the center almost always takes.

Route 2 — via the shelf (dark). Instead of dropping straight down, the center crosses over into an intermediate pair of states — the “shelf” in Figs. 3 and 5 — where it sits for a couple of hundred nanoseconds before quietly dropping to the ground state. On this route it emits no photon in the band our detector watches. As far as the detector is concerned, the center went dark.



The branching is spin dependent: a center in $m_s = \pm 1$ takes the right-hand route several times more often than one in $m_s = 0$ — so $m_s = 0$ is bright and $m_s = \pm 1$ is dim.

Figure 5: The branching that makes NV^- centers work. Both routes start from the same excited state, but their probabilities depend on the spin setting the center started in. Because $m_s = \pm 1$ takes the dark route more often, $m_s = 0$ is bright and $m_s = \pm 1$ is dim. Note also where the dark route ends: in $m_s = 0$, whatever the starting setting was.

Now the decisive fact, and the reason NV^- centers work at all.

The chance of taking the dark detour depends on which rung the center is on. A center in $m_s = \pm 1$ is several times more likely to cross onto the shelf than one in $m_s = 0$. As a result, $m_s = 0$ is bright and $m_s = \pm 1$ is dim: the spin state — invisible, magnetic, internal — has been converted into something a photodiode can measure, namely brightness. That conversion is the whole trick.

There is a bonus hiding in Route 2. Notice where the shelf deposits the center: (preferentially) in $m_s = 0$, regardless of which rung it started on. So every time an NV^- takes the dark detour, it gets sorted into $m_s = 0$.

Run the green laser for a few microseconds and the center undergoes many ups and downs, excitation and relaxation cycles. Any center that wanders into $m_s = \pm 1$ has repeated chances to fall onto the shelf, and the shelf always returns it to $m_s = 0$. Centers already in $m_s = 0$ mostly stay there. The population piles up in one rung.

Shining green light on the diamond does two jobs at once: it makes the centers glow so we can read them, and it initializes them into $m_s = 0$. Before every measurement we flash the laser and know, without having measured anything, what state the sensor is in — this is optical pumping, and a sensor that can be reset to a known state on demand is far easier to work with than one that cannot.

At this point we have a system we can set to a known state with light and read out with

light. What is missing is anything that makes it respond to the outside world. That is what the microwaves are for.

7 What the microwaves do

We have a ladder whose bottom rung ($m_s = 0$) is bright and whose upper rung ($m_s = \pm 1$) is dim, and we have just pushed every center onto the bright rung. To move them up in the *spin* ladder (this is not excitation, that is done by the green laser) we must supply exactly the energy of the gap — no more, no less. That is the universal rule for systems with discrete energy levels, and it is why spectroscopy works at all.

How much energy is the gap? It is conventional, and convenient, to quote it as a *frequency* rather than an energy, using the relation every photon obeys,

$$E = hf, \quad h = 6.626 \times 10^{-34} \text{ J s} = 4.136 \times 10^{-15} \text{ eV s}$$

It is important at this point to re-iterate that microwave drives

$$|m_s = 0\rangle \leftrightarrow |m_s = \pm 1\rangle.$$

, and the green light drives

$$|^3A_2\rangle \xrightarrow{\text{green laser}} |^3E\rangle,$$

They serve different purposes, and knowing the distinction is important. For the NV^- ground state, the gap corresponds to $f = 2.87 \text{ GHz}$. That frequency lies in the *microwave* band — the same neighborhood as Wi-Fi (2.4 GHz) and microwave ovens (2.45 GHz) — so no exotic source is needed. An antenna, a wire loop, or a strip of copper laid next to the diamond will do.

Sweep a microwave generator across a range of frequencies while the green laser runs and the detector watches the red glow:

- **Off resonance** (say 2.80 GHz): the microwaves carry the wrong quantum of energy, and the centers ignore them. Most stays in $m_s = 0$, and the diamond is at full brightness.
- **On resonance** (2.87 GHz): the microwaves now match the gap exactly and drive centers from $m_s = 0$ up to $m_s = \pm 1$. Those centers are on the dim rung — more of them take the dark route — and the total red output *drops*.

The technique has a name that spells out its own recipe: *optically detected magnetic resonance* (ODMR). It is magnetic resonance, in that microwaves drive transitions between spin states; but rather than detecting the microwaves themselves, which is hard for small samples, we detect the *optical* consequence, which is easy.

8 The graph: fluorescence versus microwave frequency

Plot the red brightness against the microwave frequency and you get a flat line with a dip in it. The dip sits at 2.87 GHz, and its depth — the *contrast* — is typically a few percent, up to about 20% in good samples.

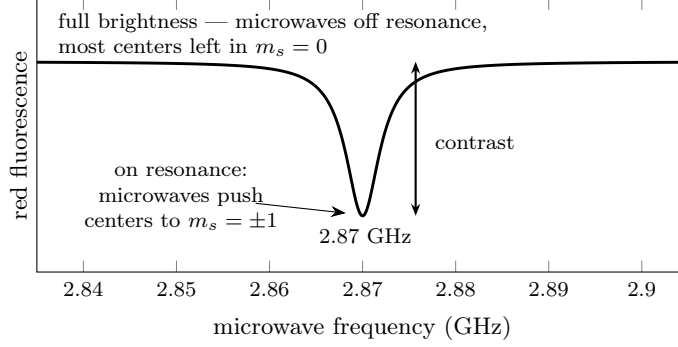


Figure 6: An ODMR spectrum with no applied magnetic field (zero-field splitting): one dip, at 2.87 GHz. Note what the vertical axis is — brightness of red light, not anything magnetic. The magnetism will be encoded in the *horizontal position* of the feature.

It is worth pausing on what kind of measurement this is. We are not measuring a force, a voltage, or a deflection. We are measuring a *frequency*, and frequency is the quantity physics measures better than any other. Every result that follows inherits that advantage.

9 Reading the graph: what the position and the width mean

9.1 The position: an energy, via $E = hf$

The horizontal position of the dip is a direct readout of the energy gap between the rungs, through $E = hf$. Putting in the number,

$$E = hf = (4.136 \times 10^{-15} \text{ eV s})(2.87 \times 10^9 \text{ s}^{-1}) = 1.19 \times 10^{-5} \text{ eV} = 11.9 \text{ } \mu\text{eV}.$$

That is a very small energy, and it is worth seeing how small by comparison.

Quantity	Frequency	Energy	Comment
Green laser photon	564 THz	2.33 eV	what we send in
Red fluorescence photon	~ 430 THz	~ 1.77 eV	what we detect
Room-temperature thermal energy	—	25.9 meV	the ambient jostling
NV ⁻ zero-field splitting	2.87 GHz	11.9 μeV	the gap we drive
Typical linewidth	~ 1 MHz	~ 4 neV	the sharpness of the dip
Splitting in the Earth's field	2.8 MHz	12 neV	the signal we want

Two things stand out. First, the optical energies are about 200,000 times larger than the microwave gap which is why Fig. 3 cannot be drawn to scale, and why cheap visible optics can read out a microwave-scale energy. Second, the gap we are measuring is about 2000 times *smaller* than the thermal energy sloshing around the room. The room is not gentle with this system; the measurement survives because we reset the spins optically many thousands of times per second and read them out before the environment has time to scramble them.

9.2 The width: how sharply the energy is defined

The dip is not infinitely narrow. It has a width — typically a few MHz for a crystal containing many NV⁻ centers — and that width is physically meaningful, not merely instrumental sloppiness.

Again through $E = hf$, a spread in frequency is a spread in energy:

$$\Delta E = h \Delta f \implies \Delta f = 1 \text{ MHz} \Rightarrow \Delta E \approx 4 \text{ neV}.$$

So the linewidth means that the gap between the rungs is not one perfectly sharp number, but 2.87 GHz give or take a few MHz. There are two independent reasons for that, and both are worth mentioning to an audience.

Not every NV^- sees the same world. A real crystal contains millions of NV^- centers, each with its own microscopic neighborhood: a nitrogen atom that failed to become an NV^- sitting a few nanometers away, a ^{13}C nucleus with its own tiny magnetic moment, a little local strain in the lattice. Each nudges the gap slightly. We measure all the centers at once, so we see the sum of their slightly different dips, which is broader than any one of them.

Short-lived signals have fuzzy frequencies. This is not special to quantum mechanics; it is true of any wave. Play a pure tone for a long time and its pitch is sharply defined; play it for a few thousandths of a second and it becomes a click with no well-defined pitch. The NV^- spins stay coherent for only a short time before their neighbors scramble them, and a signal lasting a time T can only define a frequency to about $1/T$. A microsecond gives roughly a megahertz — the numbers match, which is a hint that this is the same story as the first, told in time rather than in frequency.

The dip's position carries the signal; the dip's width sets how small a shift in position you can notice. The width is the tick spacing on the ruler. A large part of modern NV^- research — purer crystals, isotopically engineered carbon, clever microwave pulse sequences — is devoted to nothing but making that dip narrower.

10 Turning on a magnetic field: one dip becomes two

Everything so far happened with no applied field. Recall the loose end from Section 4: with no field, $m_s = +1$ and $m_s = -1$ have exactly the same energy. So when we swept the microwaves, both transitions — $0 \rightarrow +1$ and $0 \rightarrow -1$ — occurred at the same frequency and produced a single dip. Two transitions were hiding inside one feature.

A magnetic field breaks that tie. This is the *Zeeman effect*, and it is the same physics that makes a compass needle prefer one orientation: the needle's energy depends on whether it is aligned with the field or against it. Here “both spins up” and “both spins down” are the two alignments, so the field pushes them in opposite directions.

10.1 The same statement in two lines of algebra

Write the energies in frequency units (divide throughout by h , so the numbers come out in Hz). Before the field is applied, the only term is the interaction between the two unpaired electrons, which produces the built-in splitting:

$$\frac{H}{h} = D S_z^2 \implies f(m_s=0) = 0, \quad f(m_s=+1) = f(m_s=-1) = D = 2.87 \text{ GHz}.$$

Here S_z is bookkeeping for which rung you are on (0, +1, or -1). Because it appears *squared*, +1 and -1 give the same answer — which is exactly the statement that the two states are degenerate.

Now apply a magnetic field B along the NV^- axis. It adds one term, and this one is *not* squared:

$$\frac{H}{h} = D S_z^2 + \gamma B S_z \implies f_{\pm} = D \pm \gamma B,$$

where $\gamma = g\mu_B/h = 28.0 \text{ GHz/T} = 2.80 \text{ MHz/G}$ measures how strongly this particular magnetic needle responds to a field.

Because the new term is linear in S_z , it raises $m_s = +1$ and lowers $m_s = -1$ by the same amount. The tie is broken symmetrically, the single dip becomes two, and their separation is

$$\Delta f = f_+ - f_- = 2\gamma B$$

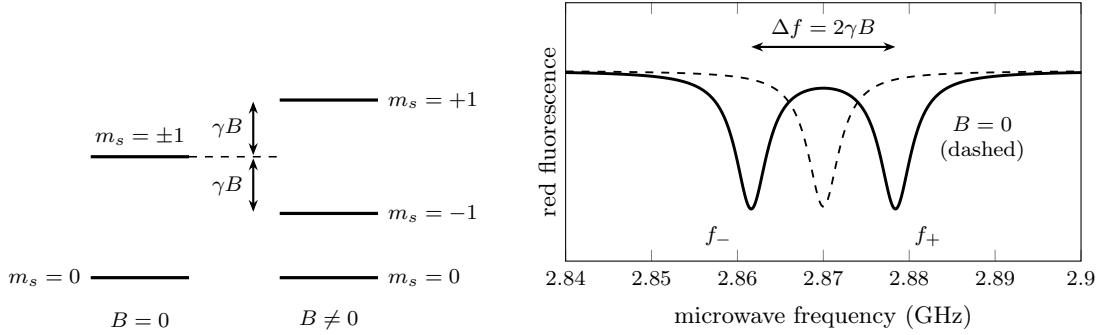


Figure 7: Left: the magnetic field raises $m_s = +1$ and lowers $m_s = -1$ by the same amount γB . Right: consequently the single dip splits into two, separated by $\Delta f = 2\gamma B$. The example shown is $B = 3 \text{ G}$ (0.3 mT), giving a 16.8 MHz splitting.

10.2 Reading a field off the graph

This is the whole instrument in one line. Measure the frequency gap between the two dips and divide by 2γ :

$$B = \frac{\Delta f}{2\gamma}, \quad 2\gamma = 5.6 \text{ MHz/G} = 56 \text{ GHz/T}.$$

Remark 1 (Worked example: the Earth's magnetic field). *The Earth's field is about $50 \mu\text{T}$ (0.5 G). Then*

$$\Delta f = 2\gamma B = (56 \text{ GHz/T})(50 \times 10^{-6} \text{ T}) = 2.8 \text{ MHz}.$$

Simply carrying the diamond outdoors splits the dip by about 2.8 MHz — comfortably measurable, and comparable to the linewidth, which is exactly why the linewidth matters. Sharpening the dip from 1 MHz to 10 kHz is a hundredfold improvement in the smallest field one can resolve.

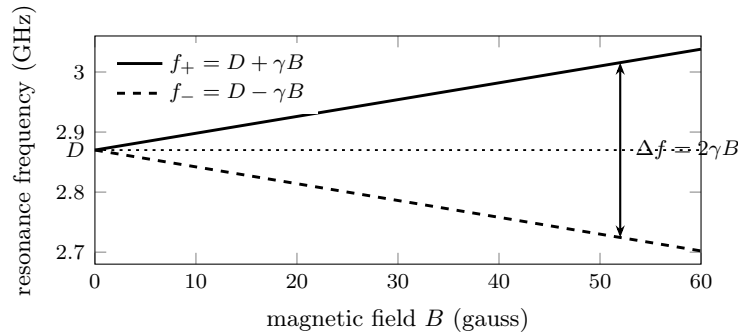


Figure 8: The two resonance frequencies fan apart linearly with the field. The slope is fixed by fundamental constants ($\gamma = g\mu_B/h$) and not by anything about the particular diamond — which is why an NV^- magnetometer needs no calibration against a reference magnet.

That last point deserves emphasis with a general audience. The conversion from “frequency gap I measured” to “magnetic field in tesla” involves only the electron’s magnetic moment and Planck’s constant. Nothing about the size of the diamond, the laser power, the detector gain, or how the device was assembled enters the answer. Build two such instruments on opposite sides of the world and they will agree, because both are counting oscillations against the same universal constants.

11 Loose ends worth knowing

11.1 The center of the pattern carries information too

The field moves the two dips apart symmetrically, so their *separation* reports the magnetic field while their *midpoint* stays at D . But D itself shifts slightly with temperature (about -74 kHz per kelvin) and with pressure and strain on the crystal. One measurement therefore yields two independent readings: the splitting is a magnetometer, the midpoint a thermometer. The same defect does both jobs at once, which is genuinely useful when the thing being measured is a living cell that must not be cooked.

11.2 Four orientations, and why that is a feature

The N–V axis can lie along any of four equivalent directions in the diamond lattice. Only the component of the magnetic field *along* a given NV’s axis shifts its levels; the sideways component barely registers. A crystal containing many NV^- centers therefore provides four differently oriented sensors in the same speck of material, producing four pairs of dips, each reporting the field projected onto its own axis.

Four projections of a three-component vector are more than enough to reconstruct the full field — magnitude *and* direction — from one measurement, with no moving parts and no blind spots.

11.3 Why many NV^- centers instead of one

A single NV center can provide nanometer-scale spatial resolution. But a single center emits only so many photons per second, and precision improves as the square root of the number of photons collected. Using N centers at once buys a factor of $\sqrt{N}$ in sensitivity at the cost of averaging the field over the volume they occupy. Which trade one wants depends on whether the goal is imaging a single protein or navigating a submarine.

12 What it is used for

- **Biology.** NV^- diamond can sit within a nanometer of a cell and detect the magnetic field produced by nerve impulses, or image magnetic particles inside tissue — at room and body temperature, a requirement superconducting sensors cannot meet.
- **Geology and planetary science.** Meteorites and ancient rocks carry a frozen record of the magnetic field in which they formed. NV^- microscopes map that record grain by grain instead of averaging over the whole sample.
- **Electronics.** Current flowing through a chip produces a magnetic field; imaging the field maps where the current actually flows, including through a fault one is trying to locate.

- **Chemistry.** The same resonance trick and underlying physics, applied to the nuclei of a nearby molecule, is nuclear magnetic resonance — the basis of MRI — but performed on volumes small enough to hold a handful of molecules rather than a test tube.
- **Navigation.** A magnetometer needing neither ultra-precise calibration nor cryogenics (though very helpful) can navigate by the Earth’s field where satellite positioning is unavailable.

13 The argument in one page

1. Diamond is a rigid, transparent, magnetically quiet crystal in which every carbon is bonded to four neighbors, with a wide (5.5 eV) energy gap.
2. Replacing one carbon with nitrogen and emptying the neighboring site gives the chain C–C–V–N–C: the **NV[−] center**. Its four dangling bonds create **localized electronic states inside the gap**, so the defect behaves like an isolated atom held rigidly in place.
3. Six electrons fill those states as $a_1^2 | a_1^2 | e_x e_y$. The first four pair off and are magnetically silent; the last two sit unpaired in e_x and e_y and combine into a **spin-1 system** with three settings: both up ($m_s = +1$), both down ($m_s = -1$), and the balanced combination ($m_s = 0$).
4. Even with no field, $m_s = 0$ lies below $m_s = \pm 1$ by $D = 2.87$ GHz, and $m_s = +1$ and $m_s = -1$ have *identical* energy.
5. **Green light** excites the center, which returns either **directly**, emitting a red photon (bright), or via the **dark singlet shelf**, emitting nothing. The dark route is far more likely from $m_s = \pm 1$, so $m_s = 0$ is bright and $m_s = \pm 1$ is dim — *spin has become brightness*.
6. The shelf preferentially deposits the center in $m_s = 0$, so the same green laser **resets** the sensor to a known state before every measurement.
7. At zero magnetic field, **microwaves at 2.87 GHz** carry exactly the energy of the gap ($E = hf$), so on resonance they move centers onto the dim rung and the red glow drops. Sweeping the frequency and plotting brightness gives **one dip at 2.87 GHz** — this is ODMR.
8. The dip’s **position** is an energy (11.9 μ eV via $E = hf$); its **width** is the spread in that energy ($\sim$ MHz, i.e. $\sim$ neV), set by how different each center’s local environment is and by how briefly the spins stay coherent.
9. A **magnetic field** adds a term linear in S_z : $H/h = DS_z^2 \rightarrow DS_z^2 + \gamma BS_z$. Being linear, it raises $m_s = +1$ and lowers $m_s = -1$ equally, breaking their tie.
10. The single dip **splits into two**, separated by $\Delta f = 2\gamma B$ with $2\gamma = 5.6$ MHz/G. Measure the gap, divide by 2γ , and you have the magnetic field in absolute units, from fundamental constants, with no calibration. (It is important to note that this is only as easy if the B and NV^- axis are aligned, otherwise we need projections.)

*Constants used: $h = 4.136 \times 10^{-15}$ eVs; $\gamma = g\mu_B/h = 28.0$ GHz/T = 2.80 MHz/G; $D = 2.87$ GHz at room temperature, drifting by -74 kHz/K; Earth’s field ≈ 50 μ T = 0.5 G. A thorough technical treatment is J. F. Barry et al., “Sensitivity optimization for NV[−]-diamond magnetometry,” *Rev. Mod. Phys.* **92**, 015004 (2020).*

References

- [1] J. F. Barry *et al.*, “Sensitivity optimization for NV[−]-diamond magnetometry,” Rev. Mod. Phys. **92**,